GGG-A-891B
June 21, 1990
SUPERSEDING
GGG-A-891A
June 3, 1954

FEDERAL SPECIFICATION

AWLS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers awls of the type commonly used by shoemakers, saddlers, cabinetmakers, etc., for piercing small holes in leather, wood, and similar materials.

1.2 Classification.

1.2.1 Types, classes, and styles. Awls shall be of the following types, classes, and styles, as specified (see 6.2):

Type I. Belt.
Type II. Bent-point.
Type III. Brad.
Type IV. Saddlers.
  Class A. Collar.
  Class B. Harness.
  Class C. Pad or seat.
  Class D. Pegging.
  Class E. Sewing and stitching, automatic.
    Style A. Enclosed spool or bobbin.
    Style B. Exposed spool or bobbin.
  Class F. Sewing and stitching, common.
Type V. Scratch.

2. APPLICABLE DOCUMENTS

2.1 The following specification and standard, of the issues in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

Federal Specification

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; Los Angeles, CA; San Francisco, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

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Military Standard

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or solicitation for offers shall apply.

American Society for Testing and Materials (ASTM) Standards:

ASTM E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

3. REQUIREMENTS

3.1 Illustration. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of awls otherwise conforming to this specification.

3.2 Dimensions and Tolerance. Unless otherwise specified in the contract or order, or in the tolerance, minimum, or maximum specified herein, all dimensions shall be in inches with an allowable tolerance of plus or minus 10 percent.

3.3 Material

3.3.1 Steel. Awl blades, needles, chucks, jaws, and wrenches shall be made of steel. The chemical composition and the heat treatment employed shall be such as to produce awls conforming to this specification.

3.3.2 Wood or Plastic. Awl handles and hafts shall be plastic or of straight-grained, seasoned hardwood, such as maple, ash, hickory, or birch. Plastic handles need not be ferruled or riveted.

3.4 Handles and Hafts. Handles and hafts shall afford a comfortable handgrip and shall have a durable coating of shellac, varnish, or other equally suitable substance which will preclude the wood from absorbing moisture. The handles only shall have a hole through the axis suitable for passage of the shank of the blades. The handles shall be either sanded or otherwise smoothly finished, to be free from slivers and chipped out pockets. The handles shall be provided with either flutes or one or two flat sides on a plane substantially parallel to the axis of the blade. Awls with one or two flat sides on the handle shall be at least 1-1/8 inches and not more than 1-9/32 inches thick (measured across the flattened sides).

3.5 Ferrules. A substantial metal ferrule shall be fitted to the blade end of the handle in such a manner that it will not become loose or detached in service.

3.6 Blades and needles. Blades and needles shall be hardened as specified herein and tempered, polished, and ready for use. Blades shall be securely fitted into the handles in order that they will not turn in the handle under service conditions.

3.7 Repair Parts. The successful bidders supplying awls covered by this specification shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification.

3.8 Identification marking. Each awl shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol, part number and country of origin of such known character that the source may be readily identifiable.

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3.9 Type I, belt awl. Belt awls shall consist essentially of a handle with a ferrule, and a blade.

3.9.1 Blade. The blade shall have a flat shank which shall pass through the center of the handle and shall be securely riveted to the handle. The opposite end of the blade shall terminate in a well-shaped point. Each side of the exposed portion of the blade shall be longitudinally grooved to a depth of 1/16 of an inch and extend to the edge of the blade, and both edges shall form a cutting edge. The blade shall contain three holes of different sizes, spaced 3/4 inch from center to center. The smallest hole shall be toward the point end of the blade, and the largest hole shall be located at the shank end.

3.9.2 Blade hardness. Blades shall show a Rockwell hardness of between 30 to 40 on the C scale at a minimum distance of 1 inch from the point.

3.9.3 Type I, belt awls shall conform to the dimensions of figure 1.

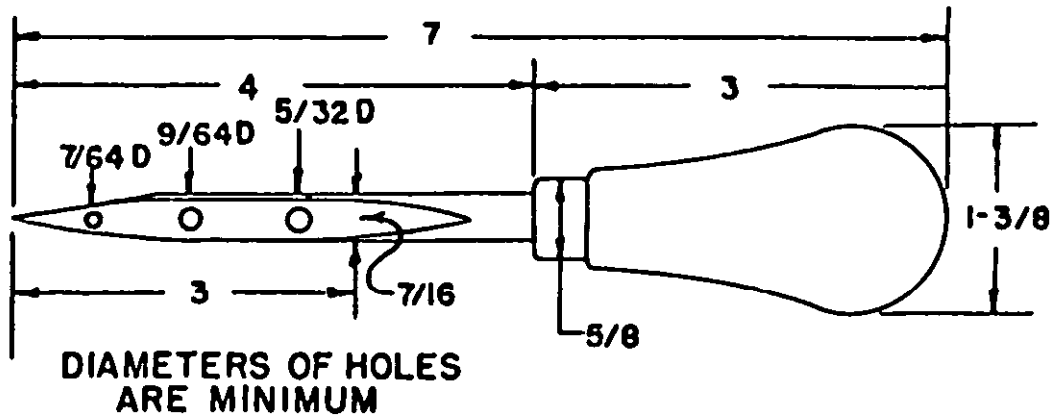


FIGURE 1. Type I, belt awl.

3.10 Type II, bent-point awl. Bent-point awls shall consist essentially of a handle with a ferrule, and a blade.

3.10.1 Blade. The blade shall be flat and bent at an angle of 15 to 20 degrees at 3/4 inch from the front end (point). The edges of the blade shall be sufficiently sharp to cut cotton twine. The edges of the bent portion shall be smooth and tapered to terminate in a sharp point.

3.10.2 Blade hardness. Blades shall show a Rockwell hardness of between 40 to 45 on the C scale at a minimum distance of 1 inch from the point.

3.10.3 Type II, bent-point awls shall conform to the dimensions of figure 2.

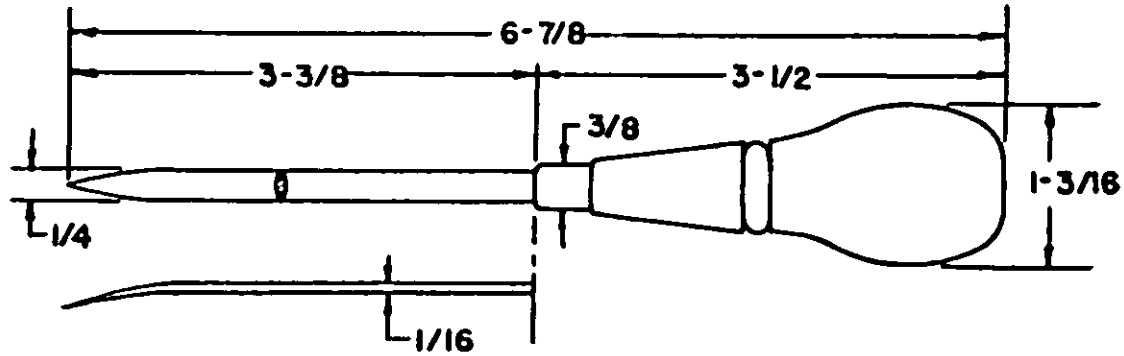


FIGURE 2. TYPE II, bent-point awl.

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3.11 Type III, brad awl. Brad awls shall consist essentially of a handle with ferrule, and a blade.

3.11.1 Blade. The blade shall have a shank which shall pass through the center of the handle and be securely riveted at the handle end. The blade shall be round, terminating in a beveled, sharp, straight edge; the bevel shall be approximately 3/16 inch long.

3.11.2 Blade hardness. Blades shall show a Rockwell hardness of between 50 to 55 on the C scale at a minimum distance of 1 inch from the point.

3.11.3 Type III, brad awls shall conform to Table I and be similar to figure 3.

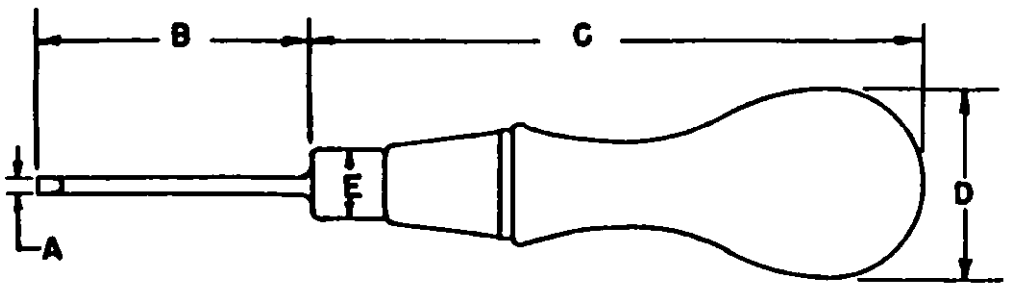


FIGURE 3. TYPE III, brad awl.

TABLE I. Type III, brad awls

Size	Length blade	Handle		Ferrule diameter, minimum
		Length	Diameter at large end	
(A)	(B)	(C)	(D)	(E)
Inches	Inches	Inches	Inches	Inch
1/16	1-1/8	3-3/4	1-1/8	1/2
3/32	1-3/8	3-3/4	1-1/8	1/2
1/8	1-1/2	3-3/4	1-1/8	1/2

3.12 Type IV, saddlers' awl.

3.12.1 Blade hardness. Type IV awls shall show a Rockwell hardness of between 40 to 45 on the C scale at a minimum distance of 1 inch from the point.

3.12.2 Class A, saddlers' collar awl. Class A awls shall consist essentially of a handle with a ferrule, and a blade.

3.12.2.1 Blade. The blade shall have a round tapered shank which shall pass through the center of the handle and be riveted over a washer at handle end. The blade end shall terminate in a well-shaped point and be flattened at a place approximately 1-3/8 inches from the point, and at the center of this flat portion there shall be an eye.

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3.12.2.2 Type IV, class A awls shall conform to the dimensions of figure 4.

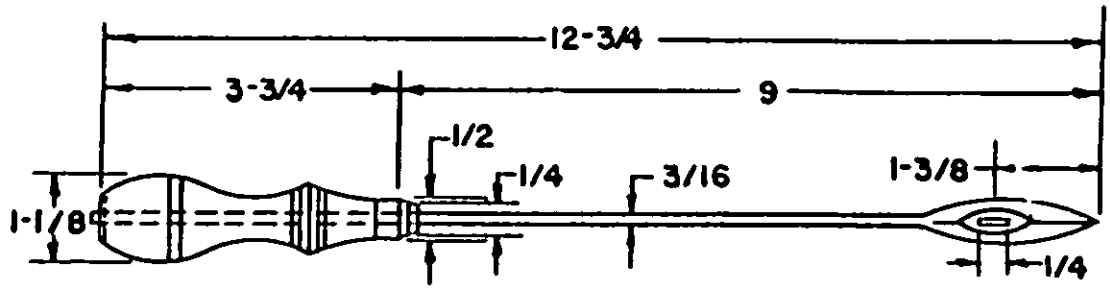


FIGURE 4. Type IV, class A, saddlers' collar awl.

3.12.3 Class B, saddlers' harness awl. Class B awls shall consist essentially of a haft, a chuck, a wrench, and six blades.

3.12.3.1 Haft. The haft shall have a threaded steel tip to engage the chuck. The steel tip shall be securely fastened to the blade end of the haft. The tip shall have a flange against which the chuck will lock. The haft shall be capped or tipped with not less than two layers of high-grade leather which shall be securely attached.

3.12.3.2 Chuck. The chuck shall be rounded in cross section, with two flat sides; it shall be internally threaded to engage the end of the haft and shall be provided with four jaws. The chuck and jaws shall be adjustable to hold the various sizes of blades as specified in 3.12.3.4.

3.12.3.3 Wrench. The wrench shall be of flat stock with an opening in one end to fit over the flat sides of the chuck; the other end shall be shaped to form a handle.

3.12.3.4 Blades. The blade shall be a diamond-shaped point in cross section, tapering for two-thirds of its length to a sharp point. The opposite end shall taper to a blunt end and shall fit in the adjustable haft. Blades shall be furnished in lengths of 1-3/4, 2, 2-1/4, 2-1/2, 2-5/8, and 2-3/4 inches.

3.12.3.5 Type IV, class B awls shall conform to the dimensions of figure 5.

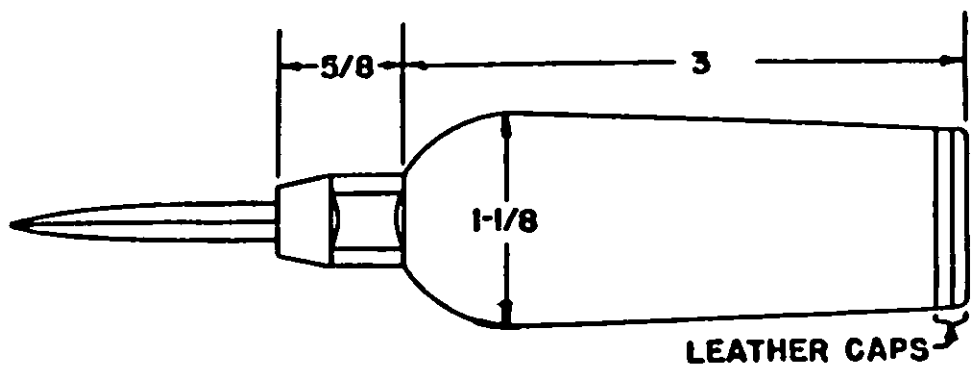


FIGURE 5. Type IV, class B, saddlers' harness awl.

3.12.4 Class C, saddlers' pad or seat awl. Class C awls shall consist essentially of a handle with a ferrule, and a blade.

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3.12.4.1 Blade. The blade shall have a shank which shall pass through the center of the handle and be securely riveted at the rear end of the handle. It shall be round and taper from the shank to a well-shaped point.

3.12.4.2 Type IV, class C awls shall conform to the dimensions of figure 6. The tolerance for the handle length shall be plus or minus 1/4 inch. The tolerance for the overall length shall be plus or minus 1/4 inch.

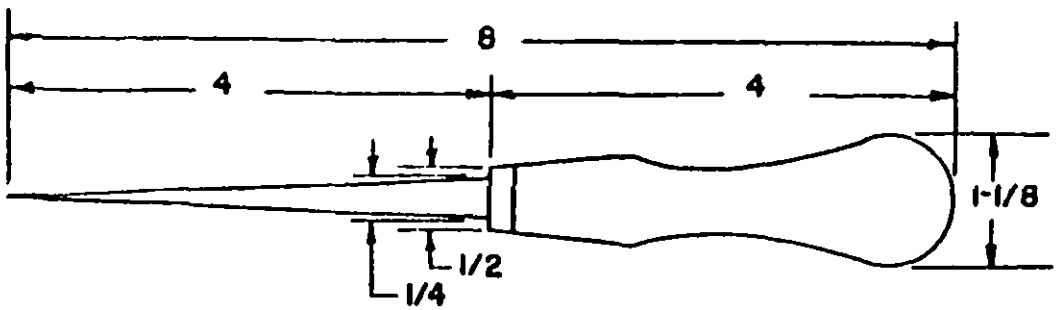


FIGURE 6. Type IV, class C, saddlers' pad or seat awl.

3.12.5 Class D, saddlers' pegging awl. Class D awls shall consist essentially of a haft, a chuck with a chuck wrench, and blades, and conform to the dimension of figure 5, except for blades.

3.12.5.1 Blades. The blades shall be of the type known commercially as "peg awl blades". They shall be square in cross section, with rounded corners, and be 1-1/2 inches in length. Each awl shall be furnished with a set of six blades.

3.12.6 Type IV, class E, saddlers' automatic sewing and stitching awl. Class E awls shall consist essentially of a handle with a chuck, and needles.

3.12.6.1 Needles. Each Type IV, Class E, Style A and B awl shall be furnished with two needles having diamond-shaped points. One needle shall have a curved point for surface sewing and the other shall be a straight needle. Each needle shall be 2 inches plus or minus 1/4 inch in length. The needles shall conform to the test requirements of 4.3.1.

3.12.6.2 Style A, enclosed spool or bobbin.

3.12.6.2.1 Handle. The handle shall have a hole approximately 1 inch in diameter by 1 inch deep in which to place the spool or bobbin. The hollowed portion shall be at the opposite end of the chuck. A steel spring cap shall be provided to cover the top of the hole and shall seat and retain the spool or bobbin in alignment. A 1/16-inch hole shall be provided from the spool or bobbin container to the outside of the handle to accomodate the thread. A tension post shall be provided on the outside of the handle. Provisions shall be made in the small end of the handle for receptacles to hold the needles. A steel ferrule shall be secured to the small end of the wood handle. The ferrule shall have three (3) holes; two shall match the holes in the handle provided for the needles, and the other shall match the thread slot. Needle holes shall be sufficiently deep to prevent contact between the needles and the chuck when tightened into place.

3.12.6.2.2 Chuck. The chuck shall be steel, round in cross-section with the jaws suitably slotted and securely attached to the handle. The chuck shall be designed to provide passage of the thread through the chuck and hold the needle securely during sewing operations.

3.12.6.2.3 Type IV, Class E, Style A awls shall conform to the dimensions of figure 7.

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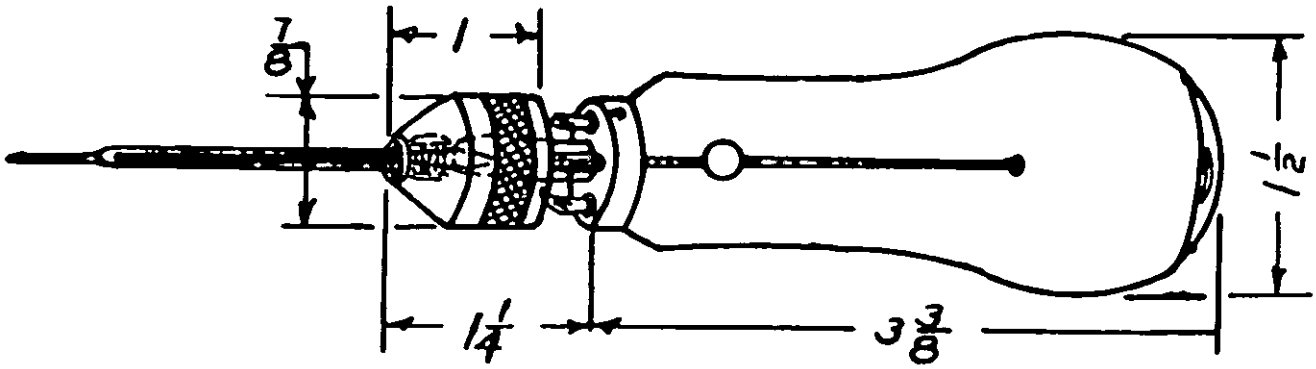


FIGURE 7. Type IV, class E, style A, saddlers' automatic sewing and stitching awl, enclosed spool or bobbin.

3.12.6.3 Style B, exposed spool or bobbin. Style B awls shall be in accordance with 3.12.6, except a chuck wrench shall be provided for tightening the chuck.

3.12.6.3.1 Handle. The handle shall be designed in a manner to have the housing supporting the spool or bobbin securely attached to the small end; the opposite end shall have an external thread to engage a wooden cap. An internal hole 5/8 inch in diameter by 1-5/8 inches deep shall be provided for holding spare needles. The wooden cap shall have an internal thread to engage the thread on the handle in a manner as to preclude any looseness or excessive play.

3.12.6.3.2 Chuck. The chuck shall be steel, round in cross-section with two flat sides. The jaws shall be suitably slotted and securely attached in the spool or bobbin housing.

3.12.6.3.3 Type IV, class E, style B awls shall conform to the dimensions of figure 8.

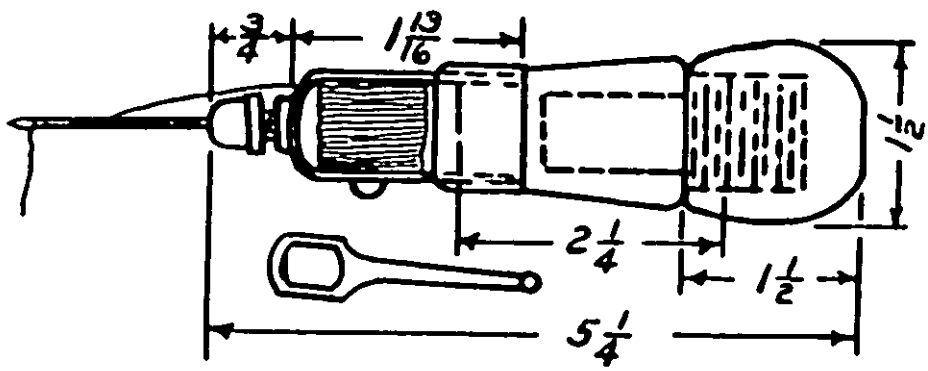


FIGURE 8. Type IV, class E, style B, saddlers' automatic sewing and stitching awl, exposed spool or bobbin.

3.12.7 Type IV, class F, saddlers' common sewing and stitching awls. Class F awls shall consist essentially of a haft with steel chuck attached, a wrench, and three needles. When specified in the contract or order, a three-inch needle (blade) shall be furnished in lieu of the three needles (see 6.2).

3.12.7.1 Haft. The haft shall have a threaded steel end to engage the chuck. The steel end shall be securely fastened to the end of the haft, and have a flange against which the chuck will lock.

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3.12.7.2 Chuck. The chuck shall be round, with two flat sides; it shall be internally threaded to engage the end of the haft and shall be provided with four steel jaws. The chuck and jaws shall be suitable for securely holding the needle or needles with which the awl is furnished.

3.12.7.3 Wrench. The wrench shall be of flat stock with an opening in one end to fit over the flat sides of the chuck; the other end shall be shaped to form a handle.

3.12.7.4 Needles. When an awl is furnished with three needles, one needle shall have a diamond-shaped point suitable for sewing, and be not less than 2 inches in overall length; the other two needles shall be ordinary straight sewing-machine needles not less than 1-5/8 inches in overall length. When the awl is furnished with the 3-inch needle (blade), the needle (blade) shall be of the bent-awl type, not less than 3 inches in overall length, not less than 3/32 inch in body diameter, tapering to an elliptical cross section toward the point, and provided with a shank suitable for engagement by the chuck. Needles shall be capable of penetrating leather of at least 3/16 inch thickness.

3.12.7.5 Type IV, class F awls shall conform to the dimensions of figure 9.

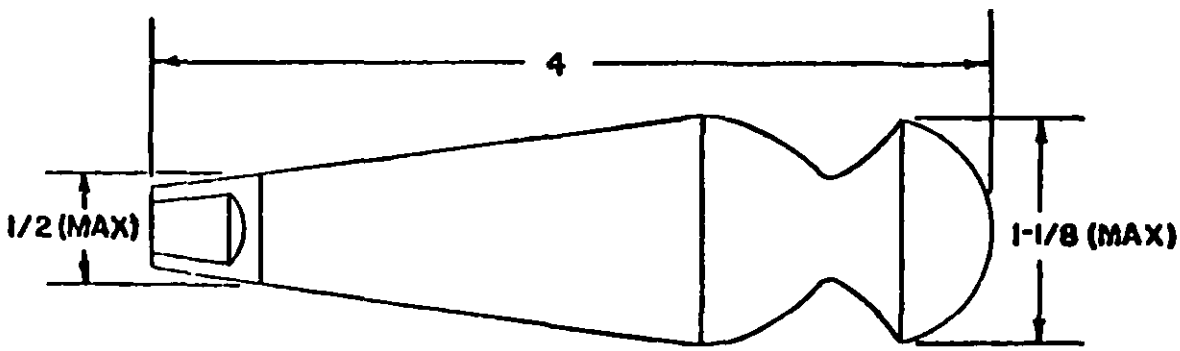


FIGURE 9. Type IV, class F, saddlers' common-sewing and stitching awl.

3.13 Type V, scratch awl. Type V, scratch awls shall consist essentially of a handle with a ferrule, and a blade.

3.13.1 Handle. The handle shall be provided with either flutes or one or two flat sides on a plane substantially parallel to the axis of the blade. Awls with one or two flat sides on the handle shall be at least 1-1/8 inches and not more than 1-9/32 inches thick (measured across the flattened sides).

3.13.2 Blade. The blade shank shall pass through the center of the handle with the head end either capped or enlarged to a diameter of at least 7/16 inch. If a cap is used, its overall thickness shall not be less than 1/4 inch. The blade shank shall be secured to the ferrule and handle by means of either a rivet or pin. If a pin is used, it shall either pass completely through the components, or a multiple fluted pin shall be forced through one wall of the ferrule and wood and through the shank of the blade. When the pin does not pass completely through the components, the top edge of the ferrule shall be either crimped or rolled into the handle. The body shall be round and gradually tapered, and terminating at the end in a sharp point.

3.13.3 Blade hardness. Type V blades shall show a Rockwell hardness of between 50 to 55 on the C scale at a minimum distance of 1 inch from the point.

3.13.4 Type V awls shall conform to the dimensions of figure 10.

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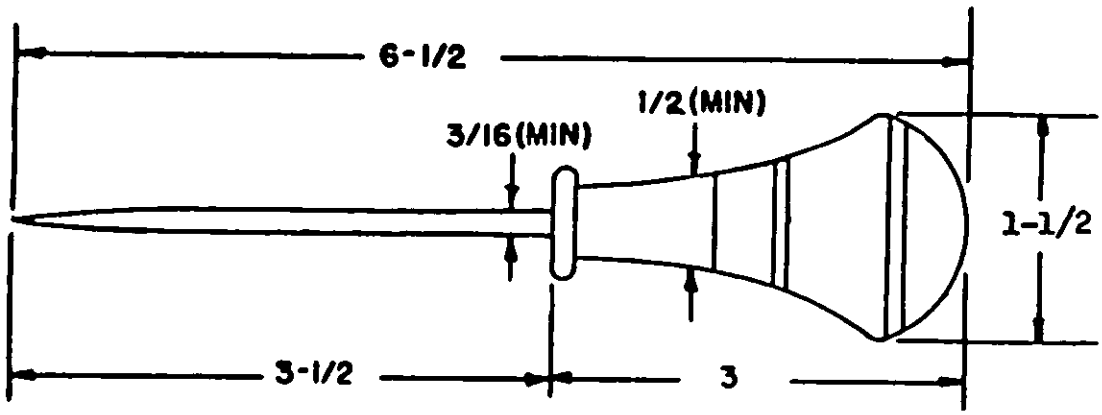


FIGURE 10. Type V, scratch awl.

3.14 Performance. Type I, II, III, and IV awls shall pass the test methods specified in 4.3.1 and 4.3.3. Type V awls shall conform to the test methods specified in 4.3.2 and 4.3.3.

3.15 Workmanship. The awls shall be free from imperfections which may affect their appearance, serviceability, safety, or durability. All parts shall be cleaned and all surfaces free of extraneous materials, fins, burrs, and rough or sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or if none, in accordance with this specification.

4.2 Examination of the end item. The lot size shall consist of all awls offered for acceptance at one time. The sample unit for this examination shall be one completely fabricated awl. Sampling shall be in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as specified in paragraphs 4.2.1, 4.2.1.1, 4.2.2, and 4.2.3.

4.2.1 Visual Inspection. Each awl shall be examined for design, material, finish construction, workmanship, and marking. The inspection level shall be level S-3 with an AQL of 1.5 percent defective for major and 4.0 percent defective for minor. Defects are listed in table II.

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Table II. Examination for visual characteristics

Examine	Defects	Major	Minor
Design	Any Characteristic not in accordance with specified requirements.....	X	
Finish	Indication of rust.....	X	
	Scratched or marred.....		X
	Handle not coated as specified.....	X	
	Touch-up not neat.....		X
	Finish tacky (not completely dry)..		X
Construction and workmanship	Part missing or not specified type or size.....	X	
	Fractured, split, bowed, malformed or otherwise impaired.....	X	
	Sharp burr or sliver.....	X	
	Operation omitted or not properly performed.....	X	
	Functioning component that requires abnormal force to operate.....	X	
	Blades and needles not properly polished.....	X	
	Blades and needles not sharpened as specified.....	X	
	Not connected or joined as specified or poorly accomplished.	X	
Marking	Missing, illegible or otherwise incorrect.....		X

4.2.1.1 Dimensional inspection. The inspection level shall be level II, with an AQL of 1.5 defects per hundred units.

4.2.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. Defects shall be scored in accordance with Table III. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

Table III. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.

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4.2.3 Testing of the end item. Each sample unit shall be tested for characteristics specified in 4.3 through 4.3.3 as applicable. The inspection level shall be S-3 with an AQL of 2.5 defects per hundred units.

4.3 Test methods.

4.3.1 Types I, II, III, and IV. Awls shall be tested by piercing 50 times through commercial belting leather of 3/16 inch thickness or commercial five ply woven fabric, rubberized, flat belting not less than 7/32 inch in thickness.

4.3.2 Type V. Awls shall be tested by scribing 50 lines of approximately 24-inch length each on stainless steel with an applied load of approximately 4 pounds on the awl. An adapter shall be used to hold the awl in an angular or perpendicular position and the test repeated by giving the awl a 90 degree, 180 degree, and 270 degree turn. The wear shall not be in excess of 0.005 inch.

4.3.3 Hardness test. Hardness of blades and needles shall be in accordance with ASTM E 18 to verify compliance with 3.9.2, 3.10.2, 3.11.2, 3.12.1, and 3.13.3.

5. PREPARATION FOR DELIVERY

5.1. Preservation and packaging. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. Awls with components (see 3.12.3, 3.12.5, 3.12.6, 3.12.6.3, and 3.12.7) shall be preserved and packaged as a unit.

6. NOTES

6.1 Intended use. The awls covered by this specification are intended for the following uses:

6.1.1 Type I. Punching and enlarging holes in leather belting.
6.1.2 Type II. Prying loose paper clips, staples, etc.
6.1.3 Type III. Piercing holes in wood and similar material, for the insertion of brads, screws, etc.
6.1.4 Type IV. Piercing holes through leather padding for chair seats, harnesses, etc.
6.1.5 Type V. Scribing lines on wood, metal, etc.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style of awls required (see 1.2).
(c) If a three-inch needle (blade) is required in lieu of three needles (see 3.12.7).
(d) Complete information covering any additional parts or accessories required.

6.2.1 When repair parts for awls are required, the following shall be specified:

(a) The manufacturer's name or trademark.
(b) Country of origin.
(c) The type, class, and style of awl.
(d) The proper name and manufacturer's number or symbol of the parts.
(e) Any other data applicable to the awl which will assist in identifying proper repair parts.

MILITARY INTERESTS:

Military Coordinating Activity

PREPARING ACTIVITY:
GSA/FSS

Custodians	Review Activity	User Activity
Army - GL	Air Force - 84	Navy - MC
Air Force - 99		

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